

UniSR - Master's Degree in Health Informatics - Data Science in Healthcare 2024-2025

Data Science Project

Project title

Myocardial infarction complications

Dataset

<https://archive.ics.uci.edu/dataset/579/myocardial+infarction+complications>

Project purpose

The aim of the project is to let you gain hands-on experience with the methods discussed in class. It is designed to deepen your understanding through practical application.

Rules

The project is voluntary and not mandatory. You may complete the project individually or in groups of up to three people.

The students who complete the project will earn up to 10 points of the exam. These points, combined with a shorter final exam, can bring the total grade up to 30 with honours. The project remains valid until the end of the exam session in February 2026.

Students who choose not to undertake the project, discontinue it midway, or decline the project grade may instead take a more comprehensive exam, which also assesses their practical skills. Through this exam, they may also achieve a final grade of 30 with honours.

Methods

You are required to use the Python language. We suggest the use of Jupyter Notebooks to keep code, notes and comments in one single file.

Methodological choices to complete the project's tasks are up to you and must be properly documented and presented in the final report.

Project phases

- **Phase 1 - Data and clinical background understanding:**
 - bibliographic search to understand the clinical background and the data available in the dataset;
 - preliminary exploratory data analysis
- **Phase 2 - Identification of a scientific question of interest:**
 - the student/group identifies a scientific question of interest, based on domain understanding
 - the question should be addressed focusing on topics treated during lectures: supervised and unsupervised approaches, model selection, model evaluation, explainability etc...
 - Identify other approaches that address the same problem and that can be compared to the selected approach
- **Phase 3 - Data preparation:**
 - load/import the dataset
 - evaluate of data quality
 - define and implement data preprocessing (e.g., management of missing values)
 - exploratory data analysis (e.g., understand data distributions)
- **Phase 4 - Methods definition:**
 - Define the methodology to complete the steps required to answer the scientific question. Specifically, at least these points should be addressed:
 - Which data (dependent/independent variables) will be used?
 - Which data transformations will be applied?
 - Which models/methods (train/test split, missing data imputation, modelling algorithm) will be used?
 - Which criteria will be used to evaluate the applied methods?
 - How other approaches that address the same problem compare to the selected approach?
- **Phase 5 - Implementation**
 - Implement the methods defined at phase 4

Important dates and deadlines

- 17/04/2025 project presentation
 - Project presentation and groups definition
- 05/05/2025 project progress review 1 – **PHASES 1 AND 2**
 - Presentation of the scientific questions by the groups to be agreed with professors/tutors
 - Presentation of background/rationale on which the scientific question is based
 - Presentation of preliminary exploratory data analysis
- 19/05/2025 project progress review 2 – **PHASES 3 AND 4**
 - Presentation of exploratory analysis results
 - Presentation of the proposed methodology and implementation strategy
 - Presentation of preliminary results
- 26/05/2025 project progress review 3 – **PHASE 5**
 - Presentation of results
- 03/06/2025 code and report due
- 04/06/2025 and 05/06/2025 project evaluation

Project progress reviews

- A presentation showing progresses should be prepared, consider the same guidelines defined in “Project evaluation”.
- Code must be available upon request.

Project evaluation

The evaluation of the projects will be focused on the correctness of the applied methodology and the clarity of reporting and presentations (project progress presentations and final presentation) rather than on models’ performance.

Punctuality in meeting the project deadlines will also be considered.

The evaluation of the project will be based on:

- **The implemented code**
 - You are required to use the Python language
 - We suggest the use of Jupyter notebooks
 - Use markdown cells to organize the code in sections and describe methodological choices directly in the notebook file
 - Code should be shared via google Colab
- **A written report**
 - The report should describe all the project phases, with explanations and justifications of all the project choices
 - The report should be structured in the following sections: Abstract, Background, Methods, Results, Discussion, Conclusions, Bibliography
 - The report can include relevant figures and tables
 - The report should be 1000-1200 words (bibliography, figures, tables, captions excluded)
- **Project presentations**
 - At the end of the course, all the project phases (1-6) will be presented
 - The final presentation should be structured in the following sections: Abstract, Background, Methods, Results, Discussion, Conclusions, Bibliography
 - The presentation must last no more than 15 minutes (maximum 15 slides plus 1 title slide).
 - It is suggested to use the UniSR presentation template. Prepare clear slides with minimal text.

Code and report must be shared with pietro.pinoli@polimi.it and a.colombo29@docenti.unisr.it by 03/06/2025.